

DEVICE FOR CHARGING Three 12V Batteries

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1 2V rechargeable batteries are used in cars, electronics projects, systems with emergency lights, electronic toys and so on. These batteries need to be kept charged to avoid damage due to self-discharge.

Circuit and working

Circuit diagram of the device for charging up to three 12V batteries is shown in Fig. 1. It is built around step-down transformer X1, two 1N5404 diodes (D1 and D2), nine 1N4007 diodes (D3 through D11), three 15V, 7815 regulators (IC1 through IC3) and a few other components.

Output voltage of a 7815 regulator can be modified with one or more

1N4007 or similar diodes to obtain the maximum voltage allowed for charging the 12V rechargeable batteries. Schottky diodes like 1N5819 (1A/40V) and 1N5822 (3A/40V) can be used instead of 1N5404 and 1N4007 diodes, respectively.

Test Points	
Test point	Details
TP0	0V (GND)
TP1	Around 20V
TP2	15V
TP3	15V
TP4	15V

or open to obtain the maximum output voltage to charge batteries BATT.1 through BATT.3. You can replace these jumpers with appropriate on/off switches of at least 2A rating.

Maximum charging current for the batteries is limited by 7815 regulators along with resistors R2, R3 and R4. Connectors CON2, CON3 and CON4 are for connecting voltmeters to measure the voltage drop across resistors R2, R3 and R4, respectively, and to calculate the charging current of the batteries.

Connect the batteries BATT.1 through BATT.3 to be charged as shown in Fig. 1. The 7815 voltage regulators can be mounted on a

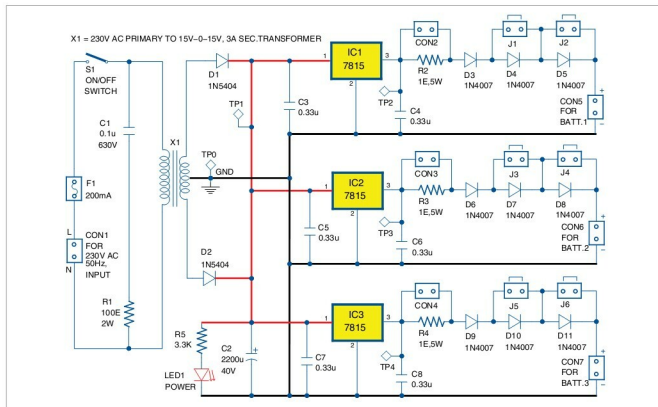


Fig. 1: Circuit diagram for the device for charging three 12V batteries